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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**

**First Semester M.Tech (MMD) (CBCS) Degree Examinations**

December 2024-January 2025

**22EVT101 – ELECTRIC AND HYBRID VEHICLES**

Duration: 3 Hours

Max. Marks: 100

**Note: Answer Five full questions choosing One full question from each Unit.**

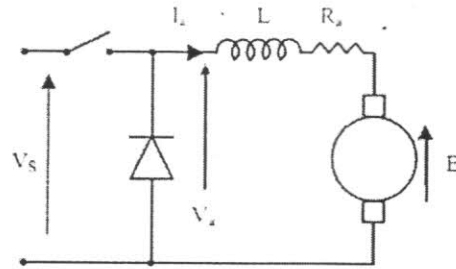
**Unit – I**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks | BT* | CO* | PO* |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) Mention the Major requirements of EV motor drive.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4     | L*3 | 1   | 2   |
| b) Explain two modes of operation in a traction motor and draw its characteristics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4     | L4  | 2   | 3   |
| c) Write the general electric vehicle configuration with a block diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4     | L3  | 1   | 3   |
| d) With the neat schematic diagrams explain the inner rotor and outer rotor in wheel configuration for an electric vehicle.                                                                                                                                                                                                                                                                                                                                                                                                                            | 8     | L2  | 2   | 3   |
| 2. a) Interpret the impact of rolling resistance and aerodynamic drag on electric vehicle performance, with relevant mathematical formula.                                                                                                                                                                                                                                                                                                                                                                                                             | 6     | L3  | 1   | 2   |
| b) List the details about the Toyota Prius HEV and compare the same with the 'Honda Insight'                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4     | L2  | 1   | 2   |
| c) An electric vehicle has the following attributes: drag coefficient $C_D = 0.25$ , vehicle cross-section $A = 2 \text{ m}^2$ , and available propulsion energy of $E_b = 20 \text{ kWh}$ ( $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$ ). Let the density of air $\rho_{\text{air}} = 1.2 \text{ kg m}^{-3}$ . Instantaneously at a vehicle speed of $120 \text{ km/h}$ , evaluate the aerodynamic drag force, power, and range, while driving in<br>(a) calm conditions with no wind and<br>(b) windy conditions with a $12 \text{ km/h}$ headwind. | 6     | L3  | 1   | 3   |
| d) State and explain the Key performance Parameters of EV.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4     | L2  | 2   | 2   |

**Unit – II**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 3. a) Analyze the distinctions between fixed gearing and variable gearing transmission systems in electric vehicles (EVs).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4  | L3 | 2 | 2 |
| b) Analyze the operational principles and evaluate the effectiveness of the four-quadrant chopper-based speed control system for a DC motor with a neat diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6  | L4 | 2 | 3 |
| c) The basic specifications for the electric motor are $P_{r(\text{rated})}$ is $80 \text{ kW}$ and $T_{r(\text{rated})}$ is $280 \text{ Nm}$ output at rated speed, a gear ratio $n_g$ is $8.19$ , and a wheel radius ( $r$ ) is $0.315 \text{ m}$ . The following assumptions are made: back emf $E_a$ of $220 \text{ V}$ at rated speed, armature resistance $R_a$ is $50 \text{ m}\Omega$ , and no-load torque $T_{nl}$ is $2 \text{ Nm}$ . Determine the armature voltage and current output by the DC-DC converter and the machine efficiency when the vehicle is operating under the following conditions:<br>a) Motoring up a hill and developing full torque at rated speed.<br>b) Cruising and developing $70 \text{ Nm}$ at the rated speed. | 10 | L3 | 2 | 3 |

4. a) A step-down chopper is used to supply power from a DC 600V overhead wire to the armature of a DC motor on a trolley bus. The armature inductance is large enough to give a smooth current and the armature resistance is  $0.4\Omega$ . The frequency of the chopper is set at 500Hz. Determine, with the aid of a circuit diagram,



- i) The ON time of the chopper to provide a motor current of 100A when the bus is stopped.
  - ii) The maximum possible generated voltage of the motor for an armature current of 100A.
  - iii. The average value of the supply current for the conditions of (a) and (b).
- b) Compare the soft-switching and hard-switching logics for electric vehicle converters.
- c) Explain the role of microcontrollers in Electric vehicle (EV) technology and outline key control strategies utilized for EV drives.

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|---|----|---|---|
| 8 | L3 | 3 | 3 |
| 6 | L3 | 2 | 2 |
| 6 | L2 | 2 | 2 |

### Unit – III

5. a) Analyze and design a comprehensive representation that elucidates the one complete cycle of the operating mode of an ARS inverter topology with a neat diagram.
  - b) Explain the configuration of a v/f controlled Induction motor drive with field weakening mode and constant torque mode.
6. a) With a neat circuit diagram, equivalent circuit, and waveforms explain two quadrant zero voltage transition converter fed DC motor drive in motoring mode.
  - b) Explain the speed control of the induction motor using a sliding mode controller and compare it with an adaptive controller.

|    |    |   |   |
|----|----|---|---|
| 10 | L4 | 3 | 5 |
| 10 | L2 | 3 | 4 |
| 10 | L2 | 3 | 5 |
| 10 | L2 | 3 | 5 |

### Unit – IV

7. a) Analyze the hybrid electric drive train system and its speed coupling within the planetary gear unit using a clear and organized functional block diagram. Provide insights into how this technology functions and the implications of its design.
  - b) Analyze and illustrate the power flow control modes employed in a series-parallel hybrid electric vehicle using a well-structured schematic diagram.
8. a) With a neat functional block diagrams explain the operating modes of the dual axle (front hybrid rear electric) complex hybrid configuration of an HEV.
  - b) Explain the different power flow control modes of a typical ICE-dominated series-parallel hybrid system with the help of block diagrams.

|    |    |   |   |
|----|----|---|---|
| 10 | L4 | 4 | 3 |
| 10 | L4 | 4 | 3 |
| 10 | L2 | 4 | 4 |
| 10 | L5 | 4 | 5 |

## Unit – V

|     |     |                                                                                                                                                                                                                                                                                                                                    |   |    |   |   |
|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 9.  | a)  | Evaluate the pack current and voltage under a 50-kW charge if the battery is fully discharged. The cell voltage drops to 2.5 V when fully discharged and an internal resistance of 2.8mΩ. A battery has 96 cells in series per string with two parallel strings. How efficient is the charging of the battery at this power level? | 6 | L4 | 5 | 2 |
|     | b)  | Discuss the points to be considered for choosing the energy storage technology for a hybrid electric vehicle                                                                                                                                                                                                                       | 4 | L3 | 5 | 4 |
|     | c)  | Explain the working principle and benefits of ultracapacitor energy storage.                                                                                                                                                                                                                                                       | 6 | L4 | 5 | 4 |
|     | d)  | Discuss in brief about various charging standards.                                                                                                                                                                                                                                                                                 | 4 | L2 | 5 | 2 |
| 10. | a)  | Explain public charging infrastructure, battery swapping station, and move and charge zone in detail.                                                                                                                                                                                                                              | 8 | L3 | 5 | 2 |
|     | b)  | With the help of neat block diagrams explain the following charging circuit configurations of EVs:                                                                                                                                                                                                                                 |   |    |   |   |
|     | i)  | microprocessor-based charger circuit                                                                                                                                                                                                                                                                                               | 8 | L4 | 5 | 4 |
|     | ii) | Arrangement of an off-board conductive charger                                                                                                                                                                                                                                                                                     |   |    |   |   |
|     | c)  | State and explain the significance of the Peukert capacity of a lead acid battery.                                                                                                                                                                                                                                                 | 4 | L3 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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